Week 12 HW- Count data

2026.05.07

1. The dataset discoveries lists the numbers of "great" inventions and scientific discoveries in each year from 1860 to 1959.

(a) Plot the discoveries over time and comment on the trend, if any.

(b) Fit a Poisson response model with a constant term. Now compute the mean number of discoveries per year. What is the relationship between this mean and the coefficient seen in the model?

(c) Use the deviance from the model to check whether the model fits the data. What does this say about whether the rate of discoveries is constant over time?

(d) Fit a Poisson response model that is quadratic in the year. Test for the significance of the quadratic term. What does this say about the presence of a trend in discovery?

Hint: poly(x,2) returns orthogonal polynomials of degree 1 to 2 over the specified set of points x. The space spanned by $\{x, x^2\}$ is the same as the space spanned by poly(x,2), but the latter eliminates multicollinearity by ensuring the predictors are orthogonal.

(e) Compute the predicted number of discoveries each year and show these predictions as a line drawn over the data. Comment on what you see.

2. The salmonella data was collected in a salmonella reverse mutagenicity assay. The predictor is the dose level of quinoline and the response is the numbers of revertant colonies of TA98 salmonella observed on each of three replicate plates.

(a) Plot the data and comment on the relationship between dose and colonies.

(b) Compute the mean and variance within each set of observations with the same dose. Plot the variance against the mean and comment on what this says about overdispersion.

(c) Fit a Poisson model with dose treated as a six-level factor. Check the deviance to determine whether this model fits the data.

(d) Fit a Poisson model what includes an overdispersion parameter. Describe the key differences in the results between this model and the one fitted in part (c).

(e) Give the predicted mean response for a dose of 500. Compute a 95% confidence interval.

(f) At what dose does the maximum predicted response occur?

(g) Fit a Negative Binomial model to the data. Use this model to repeat the analysis required in parts (e) and (f).